

EVALUATION
ONLINE LEARNING
LINKS WITH OPTIMIZATION AND GAMES
UNIVERSITÉ PARIS–SACLAY



ADAGRAD ON THE SIMPLEX AND APPLICATION TO GAMES

Let $d \geq 1$ and consider the simplex $\mathcal{X} = \Delta_d$.

- 1) Derive regret bounds for AdaGrad-Norm and AdaGrad-Diagonal in the special case of the simplex.
- 2) Let $m, n \geq 1$ be integers and $A \in \mathbb{R}^{m \times n}$. In the context of solving the two-player zero-sum game given by A , derive guarantees for AdaGrad-Norm and AdaGrad-Diagonal.
- 3) Implement AdaGrad-Norm on the simplex with the help of the following snippet which computes the Euclidean projection onto the simplex.

```
def projection_simplex(y):  
    n_features = y.shape[0]  
    z = np.sort(y)[:, -1]  
    cssv = np.cumsum(z) - 1  
    ind = np.arange(n_features) + 1  
    cond = z - cssv / ind > 0  
    rho = ind[cond] [-1]  
    theta = cssv[cond] [-1] / float(rho)
```

```
w = np.maximum(y - theta, 0)
return w
```

- 4) In the context of solving two-player zero-sum games, perform numerical experiments to compare the performance of AdaGrad-Norm with RM, RM+, the exponential weights algorithm, as well as the *optimistic* counterparts of all previous algorithms.
- 5) BONUS. — Also implement AdaGrad-Diagonal and add it to the numerical experiments.
- 6) BONUS. — Also perform numerical experiments for extensive-form games using the counterfactual regret minimization approach.

